tortoize

2026-10-14

NAME

tortoize - Calculate ramachandran z-scores

SYNOPSIS

tortoize [OPTION] input [output]

DESCRIPTION

Tortoize validates protein structure models by checking the Ramachandran plot and side-chain rotamer distributions. Quality Z-scores are given at the residue level and at the model level (ramachandran-z and torsion-z). Higher scores are better. To compare models or to describe the reliability of the model Z-scores jackknife-based standard deviations are also reported (ramachandran-jackknife-sd and torsion-jackknife-sd).

OPTIONS

The input file can be either mmCIF or PDB format and the file may be gzip compressed.

The output is a json file, if no file name is specified the output is written to *stdout*.

-h, **--help**

Display a help message and exit.

--version

Print the version and exit.

-v, **--verbose**

Increase the verbosity of the diagnostic output.

--dict=<file>

Specify a dictionary file containing restraints for residues specific to this file. This option may be specified more than once.

--log=<file>

Write a log with diagnostic information to this file. An output file must also be specified when this option is used.

REFERENCES

References:

Sobolev et al.

A Global Ramachandran Score Identifies Protein Structures with Unlikely Stereochemistry, Structure (2020), DOI: *https://doi.org/10.1016/j.str.2020.08.005*

Van Beusekom et al.

Homology-based loop modeling yields more complete crystallographic protein structures, IUCrJ (2018), DOI: *https://doi.org/10.1107/S2052252518010552*

Hooft et al.

Objectively judging the quality of a protein structure from a Ramachandran plot, CABIOS (1993), DOI: *https://doi.org/10.1093/bioinformatics/13.4.425*

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REPORTING BUGS

Report bugs at https://github.com/PDB-REDO/tortoize/issues